

The Weaker Compact-Subgroup Separation Condition: Coamenable Subgroups and Compact Ambient Quantum Groups

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Source question and status

Runde and Viselter [1, Remark 6.8] ask whether their compact quantum subgroup separation theorem remains true when a state $\mu \in C_0^u(\mathbb{G})^*$ is assumed merely not to factor through

$$\Phi : C_0^u(\mathbb{G}) \longrightarrow C^u(\mathbb{H}),$$

rather than assuming that its universal Fourier transform lies outside the image of the dual subgroup morphism. The exact question appears on printed and PDF page 19:

$$(\mu \otimes \text{id})(\mathbb{W}_{\mathbb{G}}) \cdot \hat{\Phi}(\hat{p}_{\alpha}) \in \hat{\Phi}(M(c_0(\hat{\mathbb{H}}))).$$

But $\sum_{\alpha \in \text{Irred}(\mathbb{H})} \hat{\Phi}(\hat{p}_{\alpha}) = \mathbb{1}$ strictly in $M(C_0^u(\hat{\mathbb{G}}))$ since $\sum_{\alpha \in \text{Irred}(\mathbb{H})} \hat{p}_{\alpha} = \mathbb{1}$ strictly in $M(c_0(\hat{\mathbb{H}}))$ and $\hat{\Phi}$ is nondegenerate, so we conclude that $(\mu \otimes \text{id})(\mathbb{W}_{\mathbb{G}}) \in \hat{\Phi}(M(c_0(\hat{\mathbb{H}})))$. $\square$

Remark 6.8. For $\mu \in C_0^u(\mathbb{G})_+^*$, the condition $(\mu \otimes \text{id})(\mathbb{W}_{\mathbb{G}}) \notin \hat{\Phi}(M(C_0^u(\hat{\mathbb{H}})))$ from Definition 6.4 implies that $\mu \notin \Phi^*(C_0^u(\mathbb{H})^*)$, because if $\mu = \nu \circ \Phi$ for some $\nu \in C_0^u(\mathbb{H})_+^*$, then $(\mu \otimes \text{id})(\mathbb{W}_{\mathbb{G}}) = \hat{\Phi}((\nu \otimes \text{id})(\mathbb{W}_{\mathbb{H}})) \in \hat{\Phi}(M(C_0^u(\hat{\mathbb{H}})))$ by (6.1). Moreover, if $\mathbb{G}$ is commutative, the two conditions are equivalent. We do not know whether Theorem 6.5 holds with this weaker condition as well.

6.3. Examples arising from the bicrossed product construction. A natural way to construct a

This packet gives a new affirmative answer in two broad regimes: when the compact subgroup $\mathbb{H}$ is coamenable, with arbitrary locally compact ambient $\mathbb{G}$, and when $\mathbb{G}$ itself is compact, with no coamenability assumption. The case in which $\mathbb{G}$ is noncompact and $\mathbb{H}$ is noncoamenable remains open.

Definitions and proof intuition

Write $\mathbb{W}_{\mathbb{G}}$ for the universal bicharacter, $\hat{\Phi}$ for the dual morphism, and $\hat{e}_{\mathbb{H}}$ for the dual counit. The desired weaker separation assertion says that if a state μ does not factor through Φ , then there is $\hat{\omega} \in C_0^u(\hat{\mathbb{G}})_+^*$ such that

$$\hat{\omega} \circ \hat{\Phi} = \hat{e}_{\mathbb{H}}, \quad \mu((\text{id} \otimes \hat{\omega})(\mathbb{W}_{\mathbb{G}})) \neq 1. \tag{1}$$

The key idea is to argue contrapositively. If every admissible slice in (1) has μ -value one, the proof mechanism of [1, Theorem 6.5] forces μ to absorb the Haar idempotent of $\mathbb{H}$. For a coamenable subgroup, faithful Haar averaging immediately turns this into factorization. For compact ambient $\mathbb{G}$, the polynomial coinvariant algebra replaces faithful Haar averaging: Takeuchi correspondence says that its augmentation ideal generates exactly the subgroup kernel.

The absorption lemma

Let $h_{\mathbb{H}}^u$ be the Haar state on the universal compact quantum group algebra, put

$$\vartheta = h_{\mathbb{H}}^u \circ \Phi, \quad y = (\mu \otimes \text{id})(\mathbb{W}_{\mathbb{G}}),$$

and let p be the central minimal projection corresponding to the trivial representation of $\mathbb{H}$.

Lemma 1. *If equality holds in (1) with right-hand side (1) for every admissible $\hat{\omega}$, then*

$$\mu * \vartheta = \vartheta = \vartheta * \mu. \quad (2)$$

Proof. Represent $M(C_0^u(\widehat{\mathbb{G}}))$ faithfully and put $P = \widehat{\Phi}(p)$. For every unit vector $\xi \in P\mathcal{K}$, its vector functional restricts through $\widehat{\Phi}$ to $\widehat{\epsilon}_{\mathbb{H}}$. The hypothesis gives $\langle y\xi, \xi \rangle = 1$. Since $\|y\| \leq 1$, equality in Cauchy-Schwarz gives $y\xi = \xi = y^*\xi$. Hence

$$yP = P = Py.$$

The standard subgroup-bicharacter identity gives $P = (\vartheta \otimes \text{id})(\mathbb{W}_{\mathbb{G}})$. Multiplication of universal Fourier transforms corresponds to convolution, and that transform is injective. The two displayed equalities are therefore exactly (2). $\square$

Main partial theorem

Theorem 1. *Let $\mathbb{H}$ be a compact quantum subgroup of a locally compact quantum group $\mathbb{G}$. The weaker separation assertion (1) holds, without the additional structural hypothesis (6.2) of [1], in either of the following cases:*

1. $\mathbb{H}$ is coamenable;
2. $\mathbb{G}$ is compact.

Consequently Remark 6.8 has an affirmative answer for every finite quantum group pair and for every classical compact subgroup.

Proof. Suppose, toward the contrapositive, that no $\hat{\omega}$ separates μ . By Lemma 1, (2) holds. Define the right translation coaction

$$\delta = (\text{id} \otimes \Phi)\Delta_{\mathbb{G}}^u.$$

Assume first that $\mathbb{H}$ is coamenable. If $a \in \ker \Phi$, then

$$z = (\mu \otimes \text{id})\delta(a^*a) \in C^u(\mathbb{H})_+$$

satisfies

$$h_{\mathbb{H}}^u(z) = (\mu * \vartheta)(a^*a) = \vartheta(a^*a) = h_{\mathbb{H}}^u(\Phi(a^*a)) = 0. \quad (3)$$

Coamenability identifies the universal and reduced compact quantum group algebras, so $h_{\mathbb{H}}^u$ is faithful. Thus $z = 0$. Applying the counit and using $(\text{id} \otimes \epsilon_{\mathbb{H}})\delta = \text{id}$ gives $\mu(a^*a) = 0$. Hence μ vanishes on $\ker \Phi$ and factors through Φ .

Now assume that $\mathbb{G}$ is compact. Put

$$\mathcal{P} = \text{Pol}(\mathbb{G}), \quad \mathcal{K} = \text{Pol}(\mathbb{H}), \quad \pi = \Phi|_{\mathcal{P}}, \quad \mathcal{R} = \{r \in \mathcal{P} : (\text{id} \otimes \pi)\Delta(r) = r \otimes 1\}.$$

For $r \in \mathcal{R}$, Haar averaging fixes r , while $\pi(r)$ is a right-invariant element of $\mathcal{K}$ and hence equals $\epsilon(r)1$. Therefore (2) implies

$$\mu(r) = (\mu * \vartheta)(r) = \vartheta(r) = \epsilon(r). \quad (4)$$

Let $\mathcal{R}^+ = \mathcal{R} \cap \ker \epsilon$. The Hopf $*$ -algebra $\mathcal{K}$ is cosemisimple. Thus $\mathcal{P}$ is faithfully coflat as a right $\mathcal{K}$ -comodule: coflatness follows from cosemisimplicity, and faithfulness follows because the surjection π splits as a comodule map. Takeuchi correspondence [2] now gives the exact fiber identity

$$\ker \pi = \mathcal{R}^+ \mathcal{P}. \quad (5)$$

For $r \in \mathcal{R}^+$ and $x \in \mathcal{P}$, (4) and positivity yield

$$\mu(rr^*) = \epsilon(rr^*) = 0, \text{quad}\mu(rx) = \langle \pi_\mu(x) \xi_\mu, \pi_\mu(r^*) \xi_\mu \rangle = 0. \quad (6)$$

Equations (5)–(6) show that μ vanishes on the polynomial kernel. Peter–Weyl spectral truncations preserve $\ker \Phi$ and show that $\ker \pi$ is norm dense in $\ker \Phi$. Hence μ again factors through Φ .

In both cases, failure of (1) forces factorization. The contrapositive proves the theorem. $\square$

Verification, scope, and novelty check

The proof reuses only the vector-state step before the structural hypothesis enters the proof of [1, Theorem 6.5]; the new steps are the absorption reformulation, the faithful-Haar argument, and the compact Hopf-fiber argument. The compact proof also rules out Kac–Paljutkin, Sekine, and all other finite quantum groups as counterexamples.

A bounded current search covered the exact Remark 6.8 wording, the source title and authors, compact-quantum-subgroup separation, idempotent states, and finite quantum groups. Krajczok–Skalski [3] is a later paper on positive-definite separation and cites [1], but studies approximation and von Neumann algebra separation properties rather than the subgroup factorization question. No paper explicitly proving or disproving the two statements above was found. Novelty confidence is moderate because the operator-algebraic target is specialized; mathematical confidence is high.

The unresolved regime is precisely: noncompact $\mathbb{G}$ and noncoamenable compact $\mathbb{H}$. In (3), Haar-nullity then controls only the reduced image of z , whereas the desired value is $\epsilon_{\mathbb{H}}(z) = \mu(a^*a)$ at the universal level. A general nonunital universal Hopf–Galois fiber theorem could close this gap, but it is not assumed here. Human review should focus on the Takeuchi identity (5) and the polynomial-kernel density statement.

References

- [1] V. Runde and A. Viselter, *On positive definiteness over locally compact quantum groups*, Canad. J. Math. **68** (2016), 1067–1095; arXiv:1410.1665, Theorem 6.5 and Remark 6.8, arXiv:1410.1665.
- [2] M. Takeuchi, *Quotient spaces for Hopf algebras*, Comm. Algebra **22** (1994), 2503–2523.
- [3] J. Krajczok and A. Skalski, *Separation properties for positive-definite functions on locally compact quantum groups and for associated von Neumann algebras*, Selecta Math. **31** (2025), Article 44; arXiv:2309.10046.